

2 Ugly Square Roots

If the radicand doesn't have the form $a^2 \pm u^2$ or $u^2 \pm a^2$, then you may have to complete the square before making the substitution.

2.0.1 Example:

$$\int \frac{1}{\sqrt{x^2 + 2x + 26}} dx \quad \rightarrow \quad \left(x^2 + 2x + \frac{1}{4} \right) + 26 - \frac{1}{4}$$

$$\quad \quad \quad (x+1)^2 + 25$$

$$= \int \frac{1}{\sqrt{(x+1)^2 + 25}} dx \quad \rightarrow \quad = \int \frac{5 \sec^2 \theta d\theta}{5 \sec \theta} = \int \sec \theta d\theta$$

$$x+1 = 5 \tan \theta$$

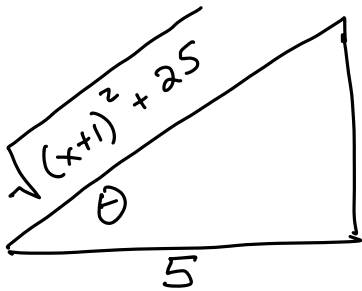
$$dx = 5 \sec^2 \theta d\theta$$

$$\sqrt{25 \tan^2 \theta + 25} = \sqrt{25(\tan^2 \theta + 1)}$$

$$= \sqrt{25 \sec^2 \theta} = 5 \sec \theta$$

$$= \ln | \sec \theta + \tan \theta | + C$$

$$= \ln \left| \frac{\sqrt{(x+1)^2 + 25}}{5} + \frac{x+1}{5} \right| + C$$



$$\tan \theta = \frac{x+1}{5}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{(x+1)^2 + 25}}{5}$$

2.0.2 Example:

Complete these squares and determine the substitution you'd use:

$$1. \sqrt{x^2 + 10x + 5} \quad \begin{matrix} \text{complete square} \\ (x^2 + 10x + \underline{25}) + 5 - \underline{25} = (x+5)^2 - 20 \end{matrix}$$

$\downarrow$
 5^2 ↗

$$x+5 = \sqrt{20} \sec \theta$$

$$2. \sqrt{x^2 + 12x + 2} \quad \begin{matrix} \text{complete square} \\ (x^2 + 12x + \underline{36}) + 2 - \underline{36} = (x+6)^2 - 34 \end{matrix}$$

$$x+6 = \sqrt{34} \sec \theta$$

$$3. \sqrt{3 + 4x - x^2} \xrightarrow{\text{negate}} x^2 - 4x - 3 \xrightarrow{\text{complete square}} (x^2 - 4x + \underline{4}) - 3 - \underline{4}$$

$$(x-2)^2 - 7 \xrightarrow{\text{negate}} 7 - (x-2)^2$$

$$\boxed{x-2 = \sqrt{7} \sin \theta}$$

$a^2 - u^2$

2.0.3 Example:

$$\int \frac{dx}{\sqrt{8x - x^2}}$$

$$(x^2 + 2x + \frac{1}{4}) + 5 - \frac{1}{4}$$

$$(x+1)^2 + 4$$

2.0.4 Example:

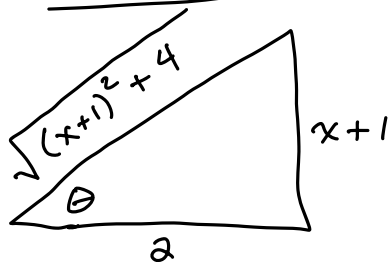
$$\int_0^2 \frac{dx}{\sqrt{x^2 + 2x + 5}} = \int_0^2 \frac{dx}{\sqrt{(x+1)^2 + 4}} = \int \frac{2\sec^2 \theta d\theta}{2\sec \theta} = \int \sec \theta d\theta$$

$$\frac{x+1}{2} = \tan \theta$$

$$dx = 2\sec^2 \theta d\theta$$

$$\sqrt{4\tan^2 \theta + 4} = \sqrt{4(\tan^2 \theta + 1)}$$

$$= 2\sec \theta$$



$$\tan \theta = \frac{x+1}{2}$$

$$= \ln |\sec \theta + \tan \theta| + C$$

$$= \ln \left| \frac{\sqrt{(x+1)^2 + 4}}{2} + \frac{x+1}{2} \right| \Bigg|_0^2$$

$$= \left[\ln \left| \frac{\sqrt{13}}{2} + \frac{3}{2} \right| - \ln \left| \frac{\sqrt{5}}{2} + \frac{1}{2} \right| \right]$$

$$7. x + 5 = 5 \sec \theta$$

$$8. x - 6 = 9 \sec \theta$$

$$9. x - 2 = 2\sqrt{2} \tan \theta$$

Worksheet 7.3

I. Pick the Trig Substitution

For the integrals below, decide which trig substitution you should use.

1. $\int \frac{3x^2}{\sqrt{x^2 + 9}} dx$

2. $\int \frac{x^3}{\sqrt{16 - x^2}} dx$

3. $\int x^3 \sqrt{9x^2 - 25} dx$

4. $\int \frac{x}{\sqrt{16 + 9x^2}} dx$

5. $\int x^3 \sqrt{5 - x^2} dx$

6. $\int \sqrt{\underbrace{(x-1)^2}_{u^2} - \underbrace{49}_{a^2}} dx$

$x-1 = 7 \sec \theta$

Complete the square, then determine the trig substitution to make:

7. $\int \sqrt{x^2 + 10x} dx$

$(x^2 + 10x + \underline{25}) + 0 - \underline{25}$

$(x+5)^2 - 25$

$x+5 = 5 \sec \theta$

$a \sin \theta \quad \sqrt{a^2 - u^2}$
 $a \sec \theta \quad \sqrt{u^2 - a^2}$
 $a \tan \theta \quad \sqrt{u^2 + a^2}$

8. $\int \sqrt{x^2 - 12x - 45} dx$

$(x^2 - 12x + \underline{36}) - 45 - \underline{36}$

$(x^2 - 12x + 36) - 81$

$(\underbrace{x-6}_{u^2})^2 - \underbrace{81}_{a^2}$

$x-6 = 9 \sec \theta$

9. $\int \sqrt{12 + 4x - x^2} dx$

$x^2 - 4x - 12 \quad \downarrow \text{negate}$

$(x^2 - 4x + \underline{4}) - 12 - \underline{4}$

$(x-2)^2 - 16 \quad \downarrow \text{negate}$

$16 - (x-2)^2$

$a^2 - u^2$

$x-2 = 4 \sin \theta$

1. $x = 3 \tan \theta$

2. $x = 4 \sin \theta$

3. $3x = 5 \sec \theta$

4. $3x = 4 \tan \theta$

5. $x = \sqrt{5} \sin \theta$

6. $x-1 = 7 \sec \theta$

7. $x+5 = 5 \sec \theta$

8. $x-6 = 9 \sec \theta$

9. $x-2 = 2\sqrt{2} \tan \theta$

II. Finish the Trig Sub

1. $\int \frac{\sqrt{16-x^2}}{x} dx$

Someone started doing this integral using the substitution $x = 4 \sin \theta$, but stopped after integrating. Put their answer back in terms of x .

$$-4 \ln |\csc \theta + \cot \theta| + 4 \cos \theta + C.$$

2. $\int \frac{1}{\sqrt{x^2-25}} dx$

Someone started doing this integral using the substitution $x = 5 \sec \theta$, but stopped after integrating. Put their answer back in terms of x .

$$\ln |\sec \theta + \tan \theta| + C$$

$$1. -4 \ln \left| \frac{x}{4 + \sqrt{16-x^2}} \right| + \sqrt{16-x^2} + C \qquad 2. \ln |x + \sqrt{x^2-25}| + C$$